



University of Management & Technology
Department of Computer Science
School of Systems and Technology
Fall Semester 2025

Instructor's Name: Syed Hamed Raza **Subject:** Deep Learning & Neural Network

Date: 19-11-2025 **Subject Code:**

Quiz #: 1 **Total Marks:** 20

Student Name: **Reg #:**

Time Allowed: 45 Minutes

Instructions for Candidates

Read the instructions carefully.

- i. Write Question number and its part number clearly.
 - ii. Attempt all questions.
 - iii. Cutting or over-writing is not acceptable
 - iv. Illustrate your answers with diagrams, graphs and tables if required.
 - v. Cross off blank pages before handing over the Answer-book to the examination staff at completion of the examination.
-

Question 1:

You are given the following training dataset for predicting Study Score (Y) from Hours Studied (X):

Record X (Hours Studied) Y (Score)		
1	1	2
2	2	5
3	3	7
4	4	10

Tasks:

- a. Compute the estimated parameters.
- b. Write the learned regression function $\hat{Y} = wX + b$.
- c. Predict the score for a student who studied $X = 5$ hours.

Solution:

Question 2:

Task: Predict whether a student will *Pass* (1) or *Fail* (0) based on the number of Practice Exercises Completed (X).

Record X (Exercises) Y (Pass=1, Fail=0)		
1	1	0
2	2	0
3	3	1
4	4	1

Assume you are training a logistic regression model:

$P(Y=1 | X) = \sigma(wX + b)$, where σ is the sigmoid function.

Tasks:

- Compute the predicted probability $P(Y=1 | X=3.5)$.
- Decide the predicted class using threshold 0.5.

Solution:

